

electricity from your body can damage the hardware components.

2. **Airflow:** Airflow plays a vital part in overclocking. A good airflow can decrease temperature drastically while bad/misaligned airflow can be fatal. Cabinets now come with at least two fans—one intake (front) and one exhaust (back). Usually, 120mm fans are preferred. Airflow is generally measured in CFM (cubic feet/metre). The more the CFM, the better the fan and its airflow. Few of the good brands are Scythe, Silverstone, Xigmatek, Antec, etc.
3. **Thermal compounds:** Also known as thermal pastes, thermal compounds are an important factor in cooling CPUs and GPUs. They transfer heat to the heat sink. A good thermal compound can reduce temperatures by 2-5 degrees Celsius. Some of the known brands are Arctic Cooling, Tuniq, etc.
4. **Cable management:** Messy cables inside the PC can hinder airflow, thus increasing the temperature. Cables hovering over the components can suffocate them resulting in their failure. So it's better to tie them, keeping them aside and away from the components' blowing/intake vents. You can use el cheapo cable ties to get the best results. If you want a permanent solution, cable covers and management kits are available off-the-shelf.



Warning: Although this guide encourages you to experiment with overclocking, I highly condemn the practice with laptops and desktops loaded with Pentium 4 (any generation) or Pentium D. These components usually run very hot and overclocking can result in system fiasco.

As I've mentioned earlier, overclocking is a vast topic, but in this article, we will only focus on CPU overclocking and not consider GPUs due to the dearth of mature

After-market heat sinks

Most hardware components suffer from the repercussions of overheating and other thermal catastrophes because of heat generation. Therefore we use heat sinks which helps in absorbing heat and keep the temperature from rising above safe levels.

To attain a higher CPU clock speed with well-controlled temperature, many companies offer after-market coolers. Most of the manufacturers allow high scalability by providing compatibility across all sockets of AMD and Intel. Therefore, even if you change the platform in future, you can continue to use the cooler, if compatible, or can increase the compatibility by spending on the mounting kit, if provided. Some of the top brands in the market are Thermalright, Prolimatech, Scythe, Arctic Cooler, CoolerMaster, etc.

If you intend to buy a cooler, then here is the list of some coolers that you can get for varying budgets:

- **Rs 1000-1,400:** OCZ Vanquishes 120mm (comes with fan installed)
- **Rs 1,500-2,000:** Thermalright Ultima 90 (doesn't include a fan) | CoolerMaster Hyper TX2 Super (comes with fan)
- **Rs 2,000-2,500:** Sunbeam Core Contact Freezer 120mm | OCZ Vendatta 2 (both include a fan)
- **Rs 3,000 and above:** Thermalright Ultra Extreme 120mm | Prolimatech Megahalems (both don't come with a fan)

overclocking software for them.

CPU overclocking varies from processor to processor. Intel and AMD follow different approaches towards overclocking. To make things a little more complicated, CPU manufacturers opt for new terminology, which is often nothing more than a gimmick. Things are not as different as they seem.

There are generally multiple ways to overclock—manufacturers provide many options to boost speed. Generally, there are two types of processors available in the market: one with the locked multiplier (refer to the sidebar titled "Jargon buster") and the other is with an unlocked multiplier. Processors with unlocked multipliers fall into a special category and are often termed as 'Special Editions'. Intel uses the term Extreme Edition and AMD's offerings are known as Black Editions.

Both these special edition processors are cherry-picked, and have the best die and unlocked multipliers in them. As usual, they come with a premium in price. Thankfully, AMD's Black Edition won't burn a hole in your

pocket and a dual core is available for under Rs 6000. On the other hand, you will have to spend around Rs 50,000 for Intel's Extreme Editions. Yes, that's half a lakh for only a processor.

Take a look at the box to become familiar with some of the jargon used in relation to CPUs.

So once you have gained some knowledge about the basic terminology used while overclocking, it's time to overclock the system.

Generally, the preferred way to overclock the system is by tweaking the BIOS settings. Refer to your motherboard manual for precise information—different motherboard manufacturers have different BIOS menu structures.

We will start by disabling some energy efficient technologies that chip giants have actuated in the chips themselves. Here is the list of options that need to be disabled:

- **AMD Cool n' Quiet:** This is a power-saving technology that lowers the CPU clock speed and voltage when the processor is idle. This keeps the CPU from